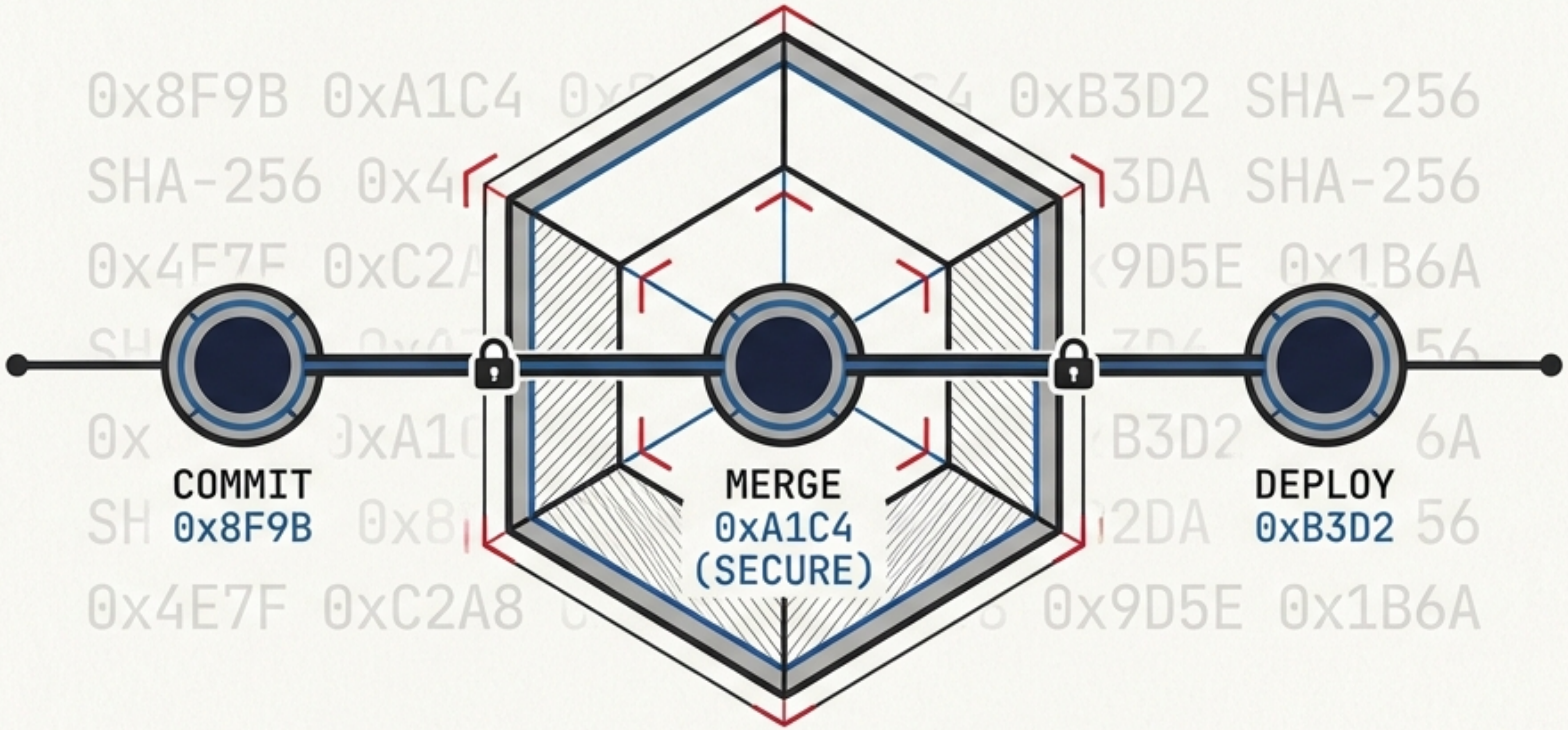
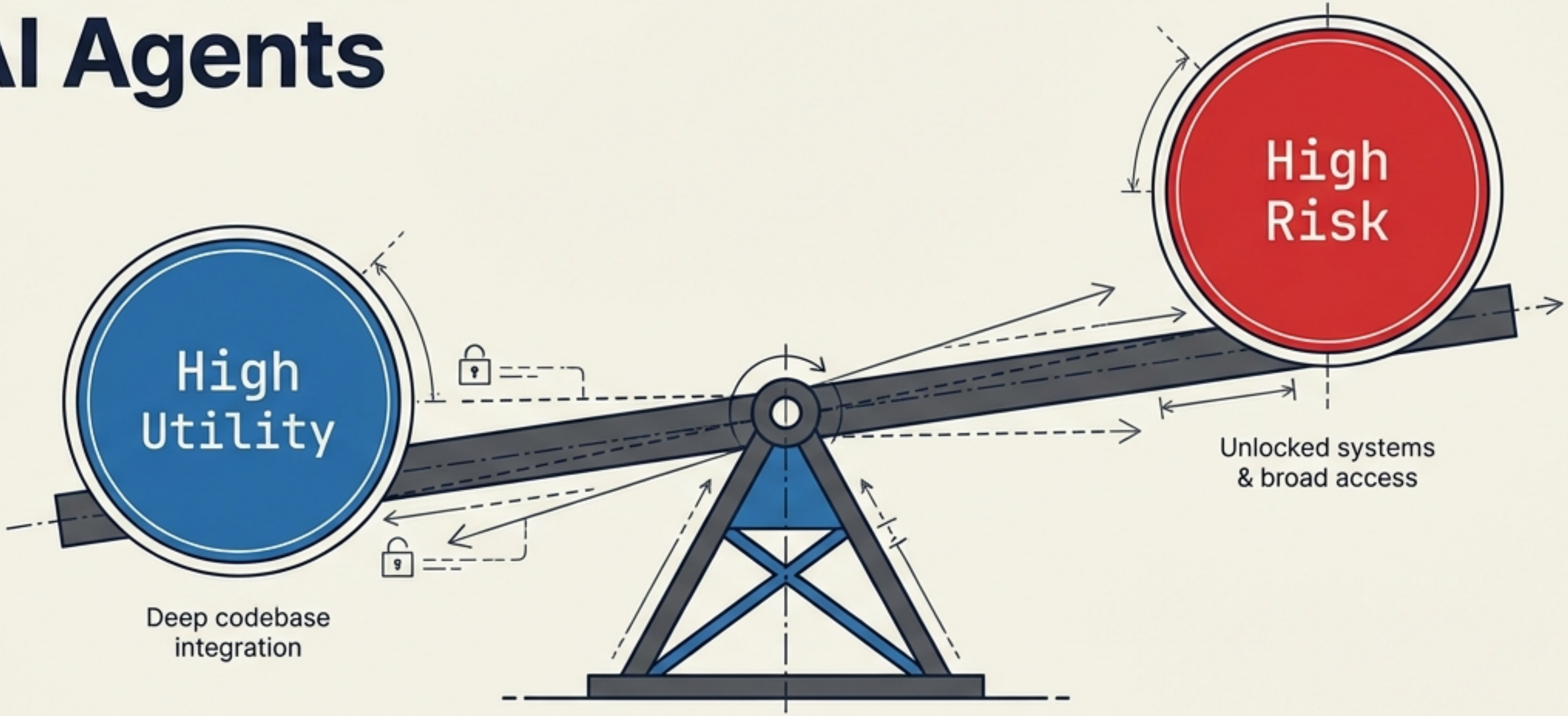
Trust Systems, Not Agents.

A mathematically secure blueprint for AI code collaboration without elevated privileges.



Based on the SG/Send Encrypted Vault Architecture Case Study.

The Core Tension of AI Agents



PARADOX STATEMENT

Every approach to AI-agent collaboration faces a fundamental paradox: an agent needs enough access to be genuinely useful, but every permission granted is a vulnerability waiting to be exploited.

The Flawed State of the Art

Methodology	Friction	Context	Security
Copy-Paste / Chat	High	Low	Safe but isolated
File Attachments	High	Lost paths	No commit history
GitHub PAT / OAuth `JetBrains Mono`	Low	High	Over-privileged risk
Local Run (Cursor, etc.) `JetBrains Mono`	Low	High	Critical risk (Runs as human)

Takeaway: Today's options force an uncomfortable tradeoff. They are either uselessly safe (chat loops) or dangerously capable (local execution).

The Danger of Identity-Based Access

Local Run Flaws

The agent runs as YOU.

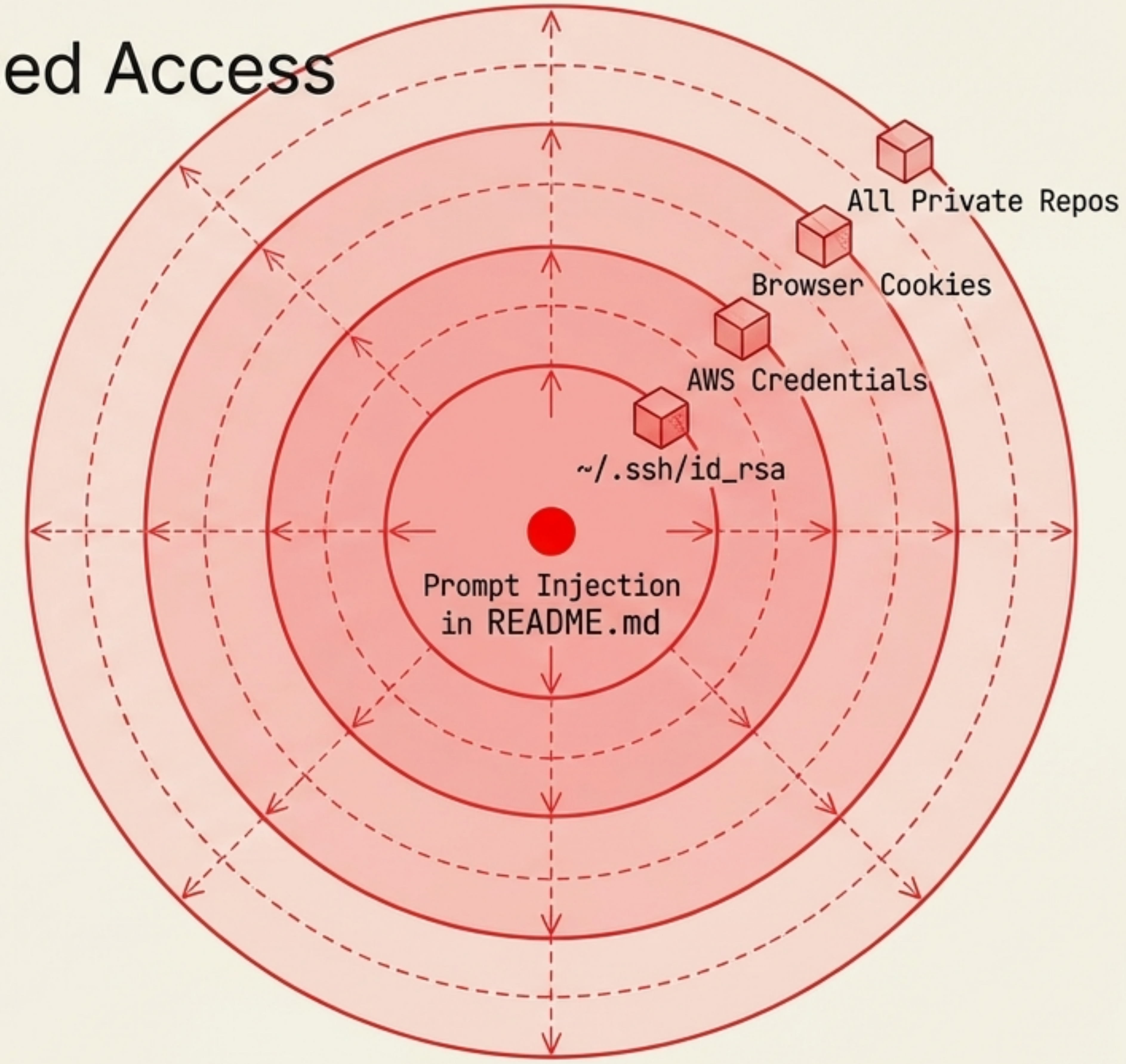
The filesystem has no privilege boundary.

A malicious instruction embedded in a downloaded artifact executes with your full credentials.

GitHub PAT Flaws

Tokens grant access to ALL repos the user can see.

Bot accounts are prohibited by ToS, meaning agents must mis-attribute commits through human accounts.



The Paradigm Shift: Identity vs. Capability

Who are you?

Broad Access. Identity inheritance grants the agent the keys to the entire system.

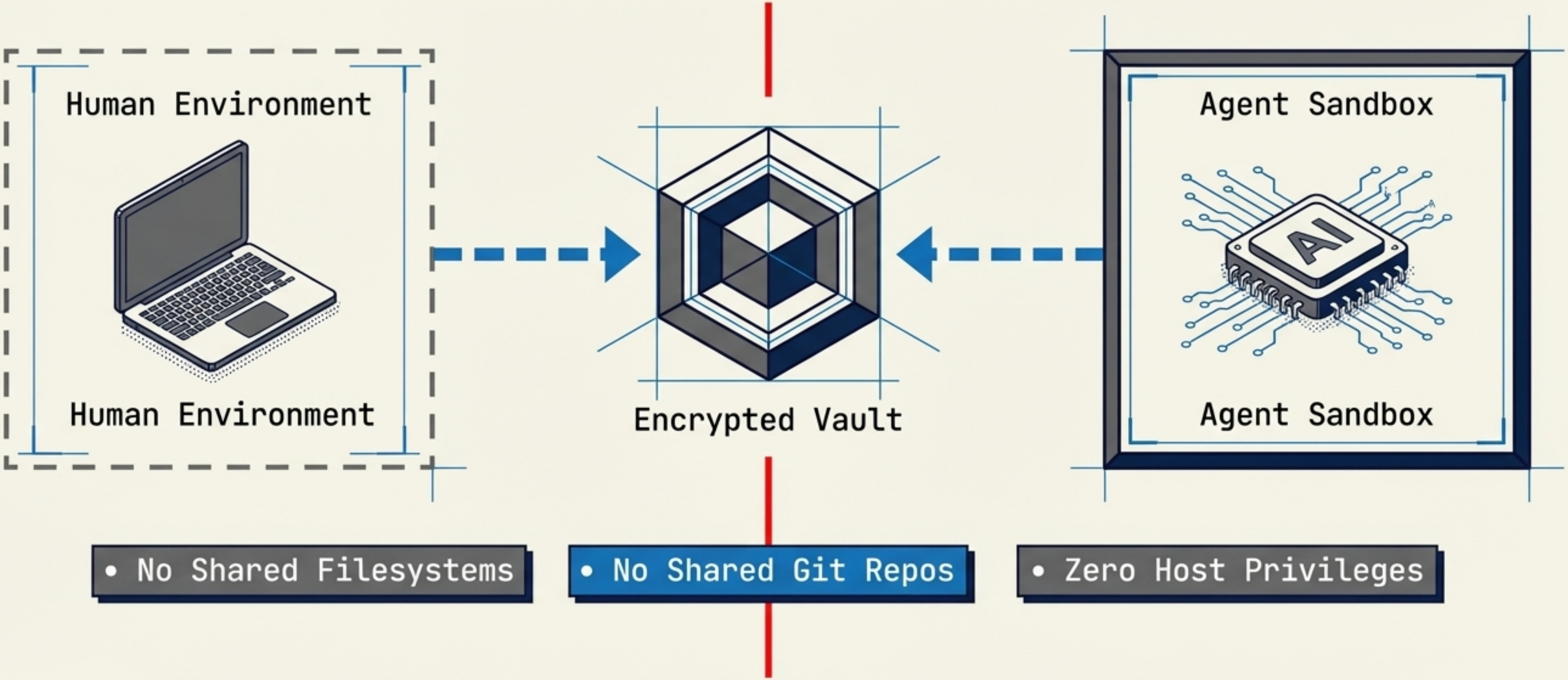
What can you do?

Exact Scope. A mathematical key that opens one specific box, with a branch that only it can write to.

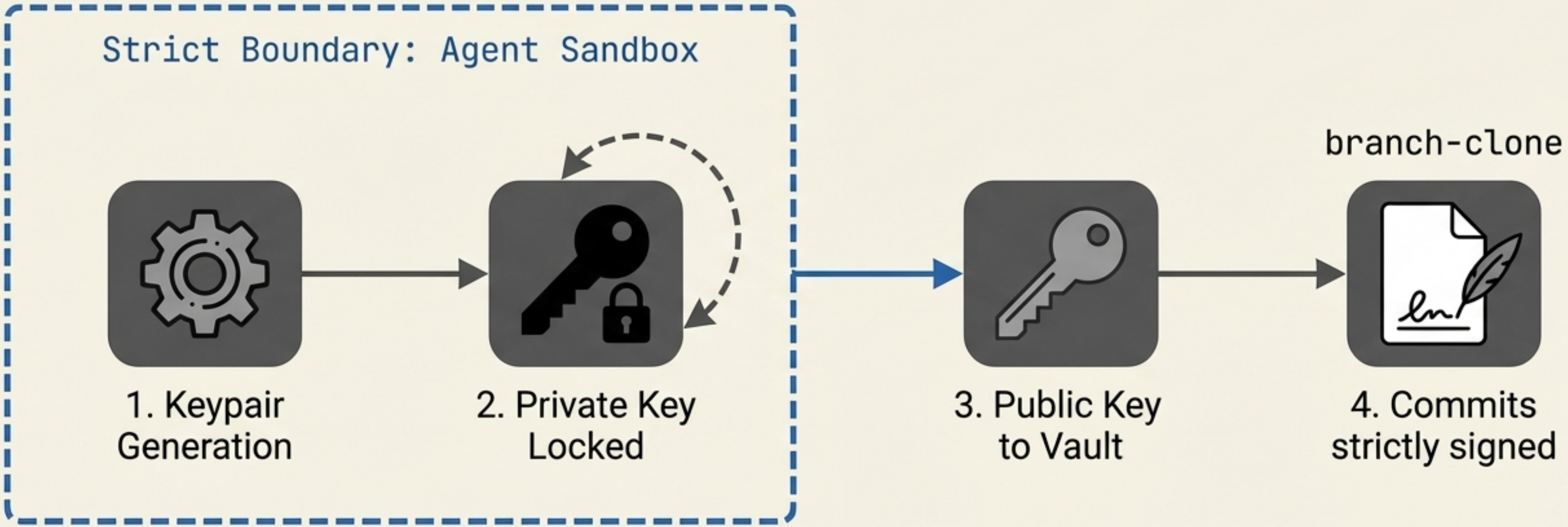
An AI agent does not need to prove who it is. It only needs a cryptographic key bound to a precise, isolated capability.

Introducing the Encrypted Vault

A secure channel without shared filesystems or elevated privileges

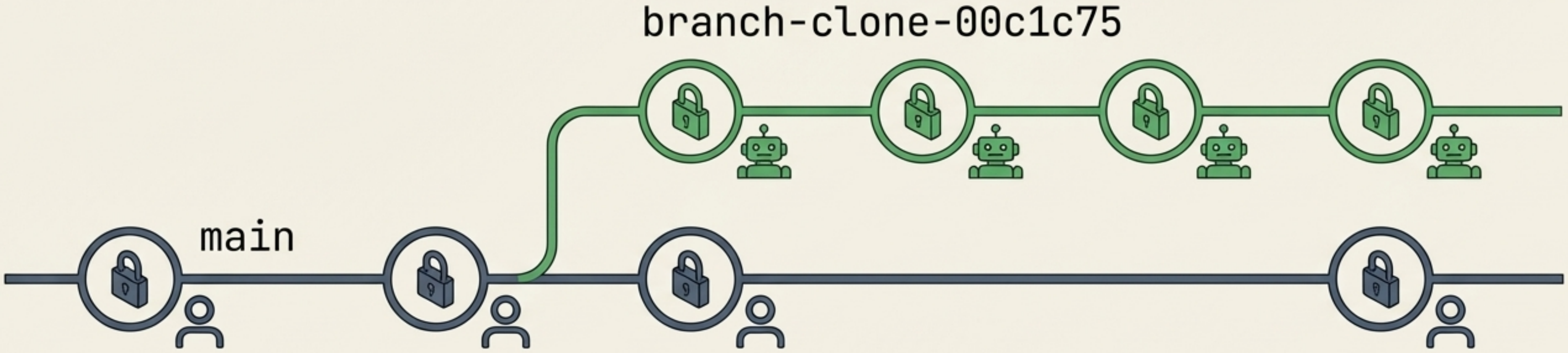


The Mechanism: Per-Branch PKI



Implication: Even if the access token leaks, an attacker cannot impersonate the clone without the private key trapped inside the sandbox.

Cryptographic Attribution Without Accounts



The agent only **commits** to its **own branch.** Every commit is cryptographically attributable to the specific clone that made it. The human controls when or if those commits merge.

Case Study Context: 19 March 2026

The Session

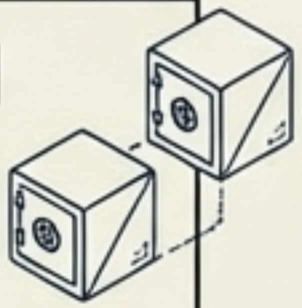
A single human developer (macOS) and an AI agent (Claude, inside a sandboxed container).

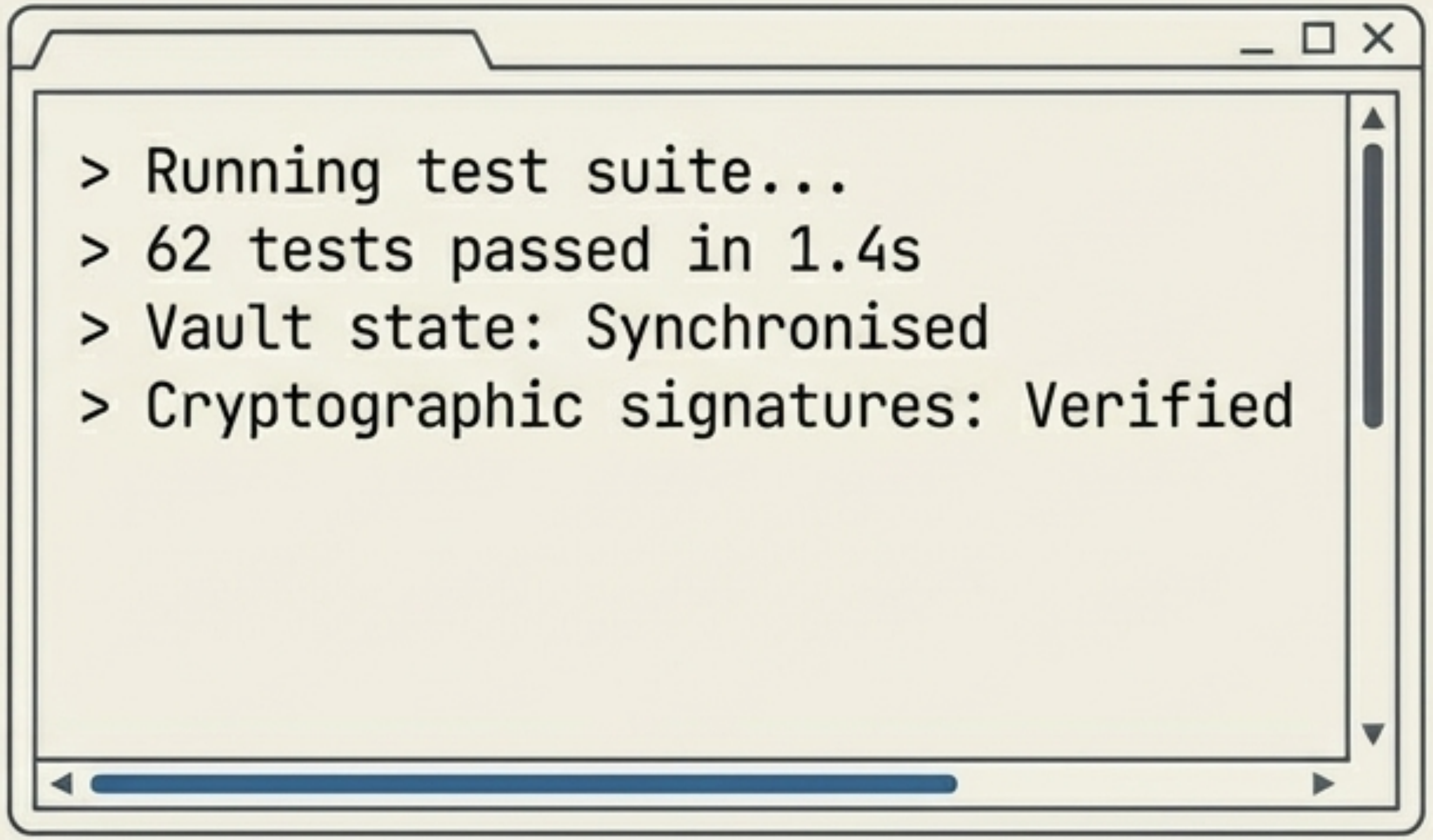
The Objective

Collaboratively build a Python CLI tool from scratch.

The Result

A fully functional architecture achieved with 62 passing tests—all facilitated through the Encrypted Vault.



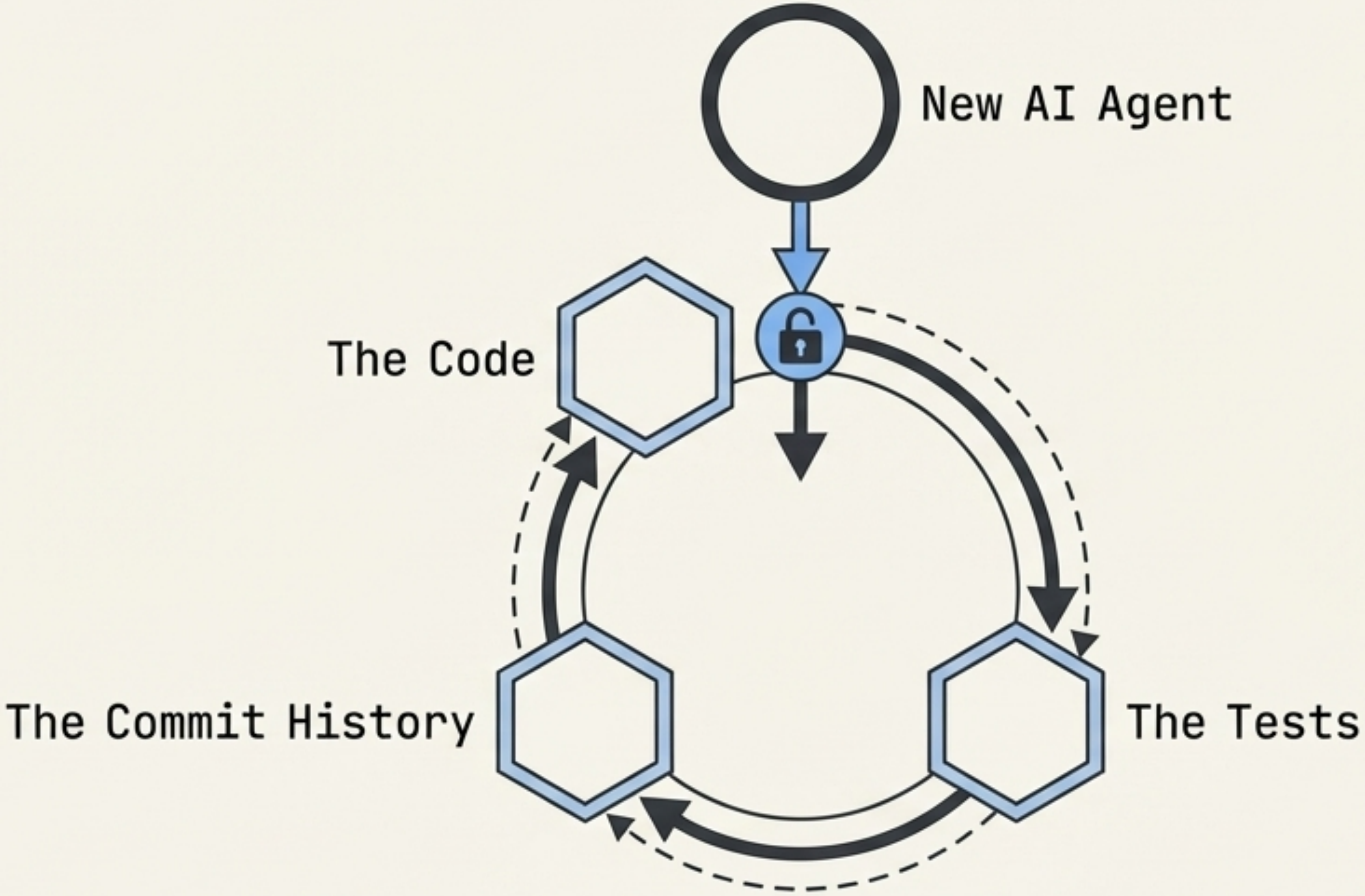


Seamless Architectural Evolution

The Pivot
Driven by human insight, the architecture evolved three times to eliminate server-side DOM manipulation.

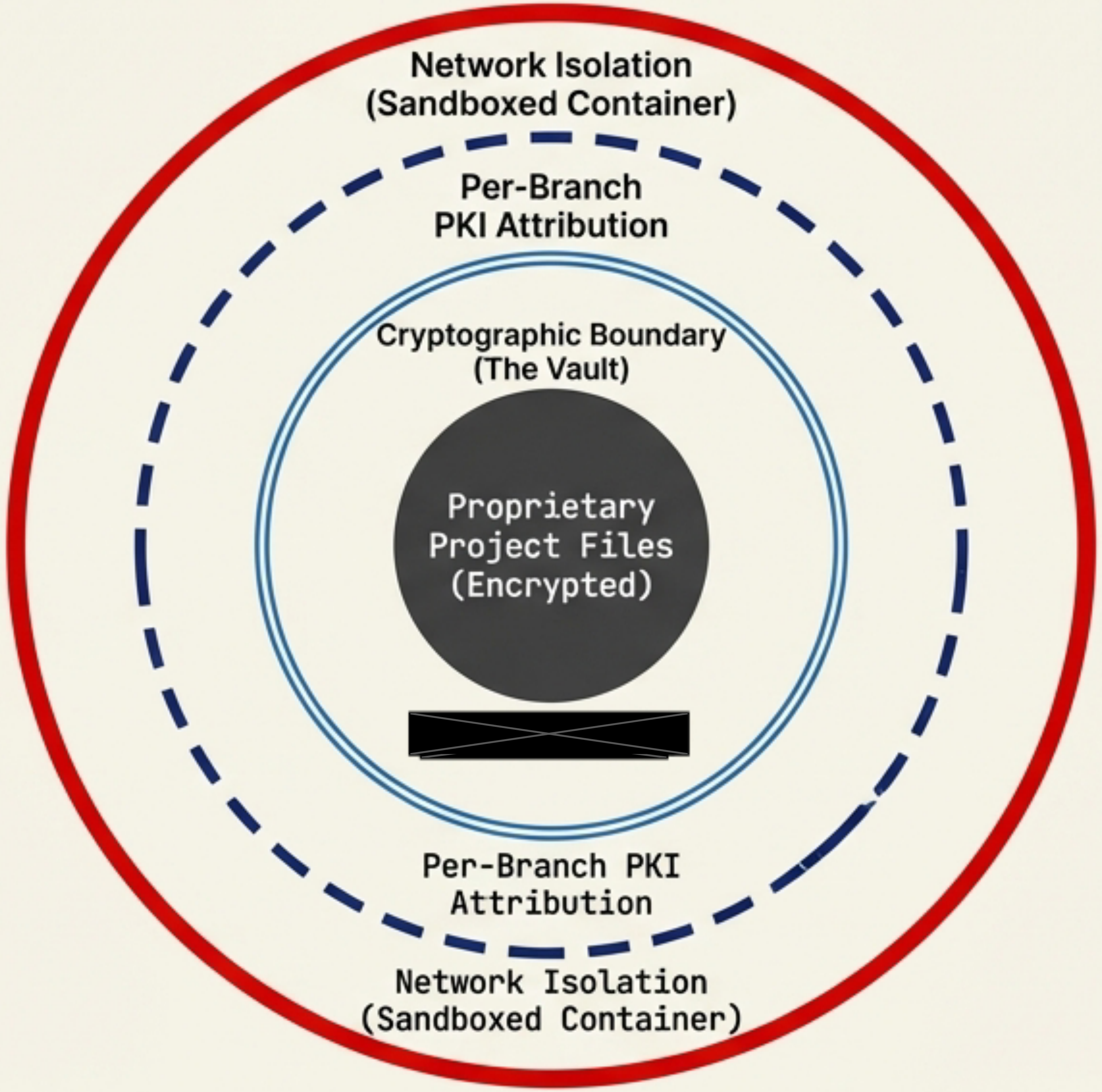
The Benefit
The vault maintained perfect state and context throughout the session without losing history or requiring manual resets.

Perfect Reproducibility: The Code is the Context



Any new agent session can pick up exactly where the last left off. No briefing documents. No manual context window stuffing. The code is the context, the tests are the specification, and the commit history is the conversation record.

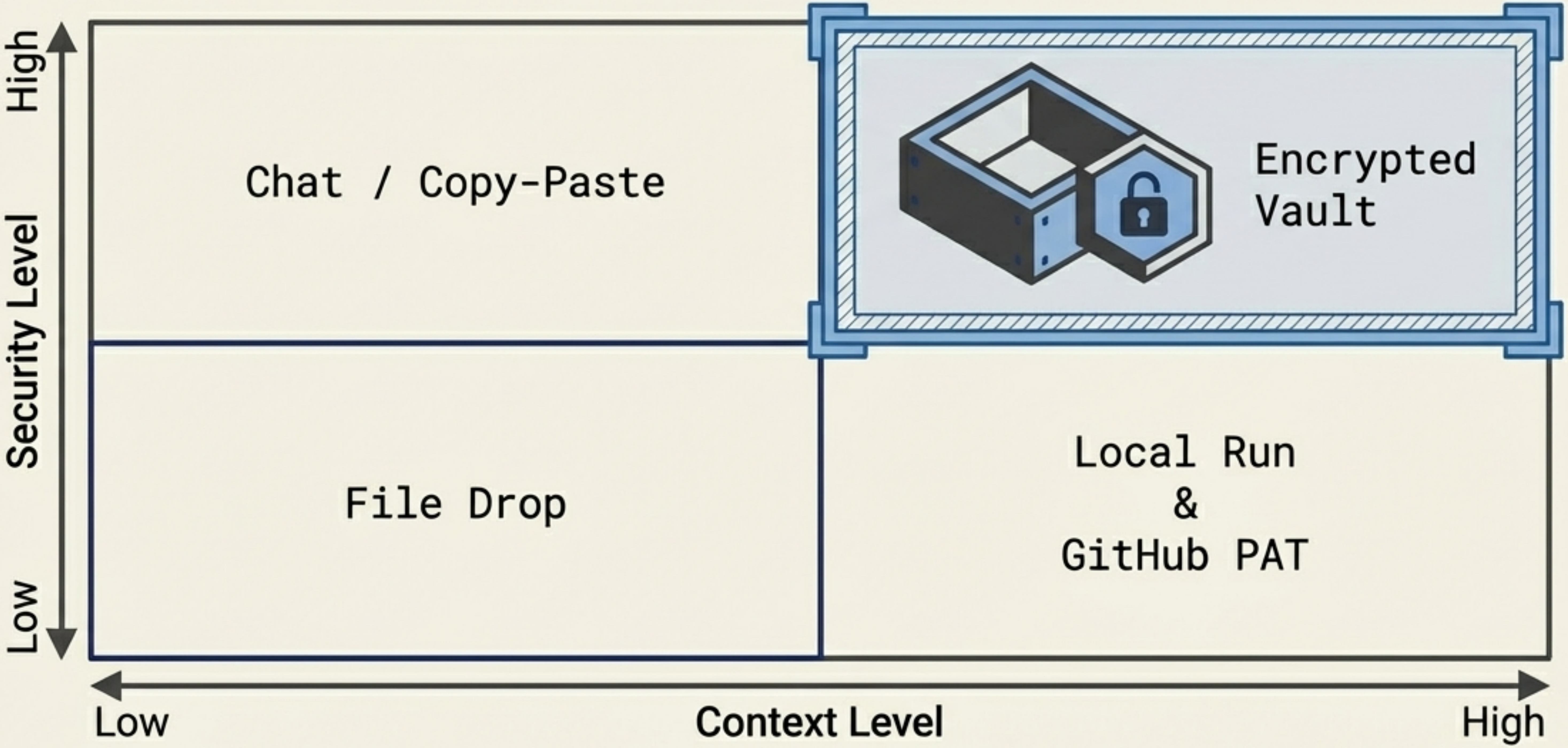
The New Security Layering



Defence in Depth

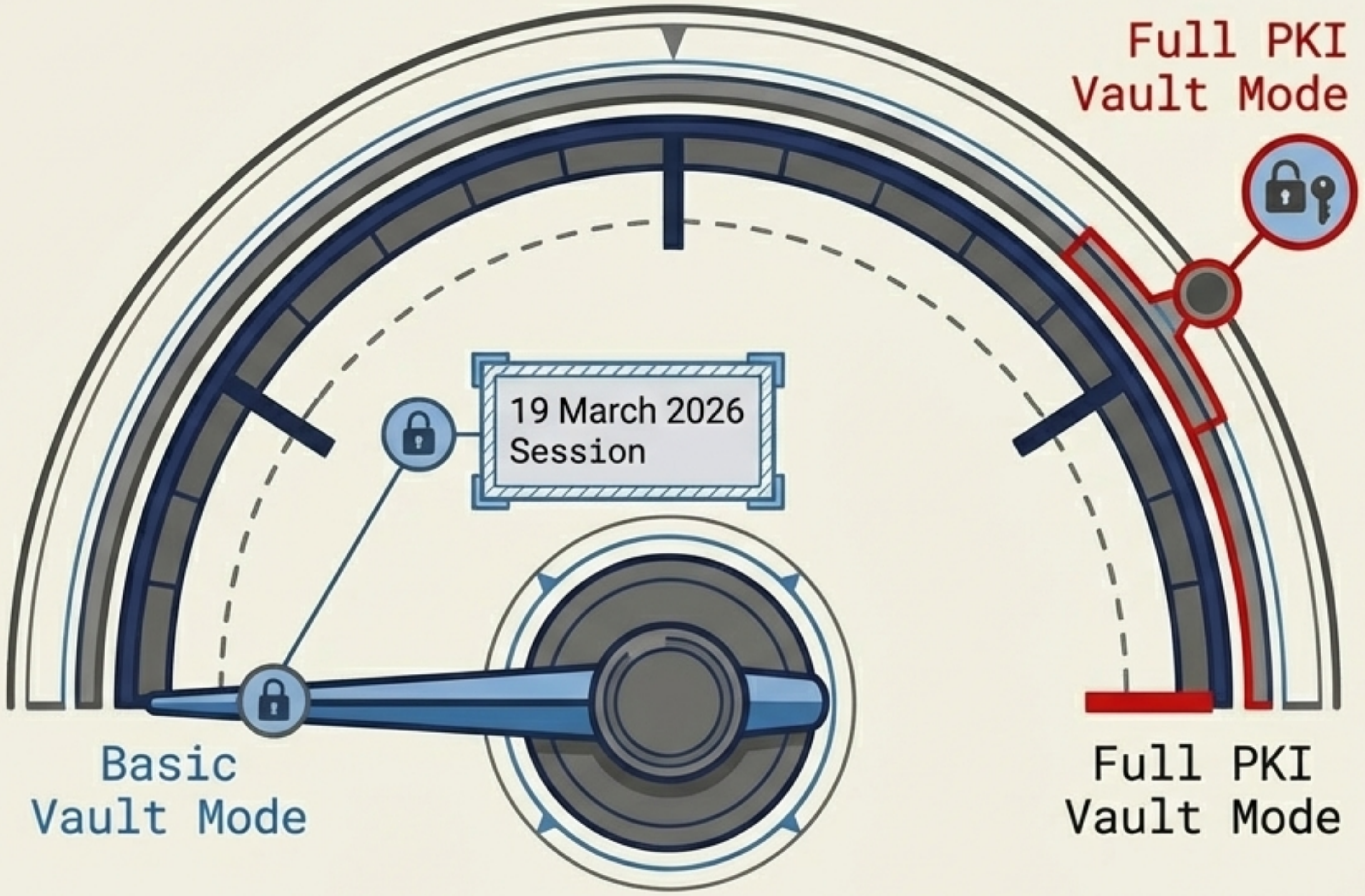
By decoupling identity from capability, capability, the system relies on physical network isolation and mathematical cryptographic boundaries, entirely bypas-ssing the need to trust the behavioural safety of an LLM.

Resolving the Tradeoff



Vaults solve the exact issues plaguing current tools, providing deep contextual awareness with a security boundary tighter than manual file transfers.

The Future: Full PKI Vault Mode



The session described utilised the `Basic Vault` operating mode—the least restrictive tier.

The architecture natively supports scaling up to `Full PKI` modes for maximum-security enterprise environments, enforcing rigid identity verification at every node.

Systems Over Trust.

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The SG/Send model provides scoped, encrypted, auditable, and revocable code collaboration. It is not about whether AI agents are trustworthy. It is about building systems where trust is simply not required.